The Merging of "Frames"

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1. Background

The word **frame**, as it appears in several co-existing traditions within the cognitive sciences (alternating with **schema** or **script**) denotes structured packages of knowledge or expectations that shape the ways in which humans enact or interpret their experiences. Frames are conceptual structures available to human beings by virtue of (a) being human and living where humans live (we know about pain, gravity, shadows and the human face), or (b) being a member of a culture (we know about marriage, school, money, and sports), or (c) knowing how to use and understand the words in our language (speakers of English know what is meant by words like *frustrate, /heartburn, sergeant* or *purple*).

2. Cognitive Frames

Though the "frames" of frame semantics (introduced below) have much in common with the frame concepts developed outside of linguistics, I shall use **cognitive frames** to refer to the structures that people invoke to make sense of their observations, and **linguistic frames** to refer to those frames that are conventionally associated with specific linguistic material. I shall say that people **invoke** frames, and that linguistic frames **evoke** frames in the mind of those who know the language.

Cognitive frames participate in reasoning (enabling us to draw inferences from partial knowledge), *expecting* (helping us know what to look for, or wait for, in the course of everyday experiencing), *remembering* (providing structures to associate things in our minds), *perceiving* (enabling us to categorize what we are perceiving), and *behaving* (interpreting the behavior or others and guiding our own).

An important claim in frame-informed Artificial Intelligence is that human reasoning uses frame knowledge in ways that are superior to reasoning based solely on logic and bare facts (Minsky 1974). Suppose, for example, that you see neatly dressed children going into someone's house carrying brightly wrapped packages. You quickly infer that what is going on is a Birthday Party. Once you have made this decision, you find networks of predictions you are prepared to make about what is going to happen inside that house when the party begins, predictions involving cake, candles, toys, games, and the dreadful birthday song.

In such a case we are making sense of bits of observed behavior—a visual experience—-by invoking our understanding of a common cultural institution. Minsky early on illustrated the phenomenon in terms of responses to a linguistic object, a particular mini-text:

(1) Mary was invited to Jack's party. She wondered if he would like a kite.

Guests at a birthday party are expected to bring gifts, a kite is a toy, and toys are appropriate gifts for children—so it's probably a children's birthday party. Although in this case the frame is invoked to interpret a piece of linguistic input, this is not yet what I mean by language-triggered evocation. The birthday party frame emerges by abductive reasoning based on what the sentence means: there are no words in that bit of text that by themselves evoke the concept. Only if the text had included expressions like *birthday party* or *birthday present* could we say that the frame was evoked because of the interpreter's knowledge of linguistic conventions.

The much-discussed Restaurant Script (Schank and Abelson 1977) and the fact that members of our culture "know" it, explains the expectations, behavior, and occasional surprises that people associate with restaurant experiences. The frames treated by Goffman (1974) show how social behaviors are interpreted as enactments of pre-established behavioral routines. Since these can be differently invoked between observer and participant, or among participants, they figure importantly in situations of misunderstanding, pretense and deceit.

Having a frame can make it possible to hold facts and experiences together in memory. The ability to see a situation as an instance of a particular frame helps select what to remember and what to ignore: if I need to remember what happened in a birthday party I attended, I am aided by my knowledge of the frame. But in certain kinds of experimental settings, for example in some work of psychologist James Jenkins (Jenkins, Wald and Pittenger 1978), framing can account for false memories. (Subjects would wrongly "remember" having seen photographs that represented sub—events of frame-construed situations.)

Since frames are introduced into an interaction by the communicator, rather than by the addressee, the initiator of a communication is free to "frame the discourse": journalists, talk-show hosts, politicians, preachers, opinion pollsters and advertisers are likely to choose framings that addressees have to confront, either by accepting them or explicitly challenging them. A journalist who uses the phrase *resistance movement* to refer to groups the nation designates as *insurgents* will be heard as making an unwelcome political statement. An opinion poll question that asks whether some potentially offensive act should be *permitted* is likely to evoke a desire not to let people "get away with" such things; a version of the question that asks whether that same act should be *forbidden* is likely to evoke the urge to resist authority. Polling subjects are likely to say *no* to both questions, although a *no* to one version entails a *yes* to the other. Political discourse in America today has a high consciousness of language-guided framing as a way of manipulating public opinion. (Lakoff 2004, Luntz 2007)

3. Linguistic Frames

In my own history as a linguist, the use of the word frame has gone through an evolution that ended up essentially merged with its use in the other cognitive sciences. Since I have often been asked about the connections between these uses, I will explain it now. (See also Fillmore 2006)

3.1. Fries Frames

In the studiedly "descriptivist" linguistics tradition in which I was first trained, the term "frame" was used to refer to templates of filled and unfilled slots that could define types of linguistic entities. In the case of word classes, the ability of a word to occur naturally and meaningfully in frame (2a) below but not (Zb) can be taken as a "test" for the class of scalar adjectives, whereas a word's welcome in frame (2b) but not (2a) could serve to identify the class of end-of-scale adjectives.

(2a) [is very __ .]
(2b) [is absolutely __ .]

The elements of such frames are just words and junctures (indications of unit boundaries). A major goal in devising such tests was to make it unnecessary to appeal to a pre-existing classification of "parts of speech": a trick like having the copula precede *very* blocks cases where *very* can be preceded by the and followed by a noun, as in the *very end*; and the full stop following the gap, indicating sentence-final position, blocks the use of *very* as modifier of an adverb, as in is very gradually changing. In this way it becomes possible to avoid any mention of the traditional categories by means of purely "objective" classifications.

The word class discovered (or validated) by frame (2a) includes *hungry, tall, large, happy*, etc.; the one discovered by frame (2b) includes *gigantic, enormous, fascinating, exhausted*, etc. Neither frame naturally welcomes such adjectives as *fiscal, binary, judicial, or national*. We find, thus, that purely distributional tests appear to be capable of reliably identifying three different subclasses of adjectives. Especially as argued by C. C. Fries (1952) and Z. Harris (1951), such tests were a necessary antidote to the inclination to use such "unscientific" criteria as grouping words according to whether they name things, properties, or events, or such assumed-prejudicial assumptions as that any language will have nouns, adjectives and verbs.

3.2. Subcategorization Frames

In the generativist tradition, the basic distributional subclasses of verbs (especially) was defined within partial phrase-structure representations in which the lexical "slot" in a sub—tree was identified in terms of the categories of the phrases that could accompany them within a single verb phrase. These **subcategorization frames** (Chomsky 1965) presupposed both a constituent-structure organization of sentential syntax and language-universal categories of phrase-types, such as NP (noun phrase), PP (prepositional phrase), VP (verb phrase) and S (sentence or clause).

This time it was not necessary to invent special-purpose frames to match the linguist's pre-existing hunches; the contexts for verbal classification were provided by the very rules that were needed for generating the underlying organization of all possible syntactic tree types in which lexical verbs figured.

The subcategorization frame for verb-headed phrases shown as [__] (where nothing accompanies the verb inside the VP) covers vanish, sleep, grow, smile; [___NP] (with a following NP) covers eat, break, see, know; [_ PP] covers depend (on), object (to), cope (1:/it/2); [_ S] covers t/1z'nk,,én022/, suggest, believe; continuing through [__ NP S] for the class including persuade, inform, tell; [__ NP PP] for place, put, insert; and so on.

3.3. Case Frames

My own early work on what came to be called case grammar (Fillmore 1968) proposed a set of quite general (presumed universal) semantic-role categories that could stand for how the dependent elements of a verb-headed sentence related to the type of situation introduced by the verb, beginning with such traditional notions as Agent (the enactor of some controlled event) and Patient (the undergoer of a change designated by the verb).

Borrowing the word case from the tradition of studies of the uses of the categories found in such "case languages" as Latin, Greek, Sanskrit or Russian, but using it to refer to semantic functions rather than categories of grammatical form, I defined case frames as the configurations of semantic cases that could constitute the argument structures of particular classes of lexical items, most saliently verbs. The full combinatory description of a verb would consist, then, of a pairing of a case frame with the manner in which the phrases representing the individual cases are realized in the syntax. (For exempla, for a simple transitive ver, we can have the pairing {agent-as-subject, patient-as-object}.)

The various arrangements of case roles could be seen as reflecting the types of situations expressible in a verb-headed sentence. Combinations of such roles could serve to define many significant classes of verbs, in terms of semantic roles rather than syntactic context, including the many cases where a given semantic role could be expressed by more than one phrase type. Whenever researchers exploring the classification of verbs definable by case frames encountered verbal meanings that could not be accommodated in the current list of cases in a straightforward way, they could either add to the existing list of cases or devise an auxiliary theory of metaphoric interpretation. Acts of Giving were situations involving an Agent, a Theme, and a Recipient: maybe acts of Showing could be accommodated to these. Acts of Removing involved an Agent, a Theme and a Source): maybe acts of Depriving could be metaphorically assimilated here. Events of spontaneous Experiencing required an Experiencer and some Content (*I remembered the accident*), whereas events of caused Experiencing could involve a Stimulus, an Experiencer and a Content (*The noise reminded me of the accident*.) or an Agent, an Experiencer and a Content (*My father reminded me of my obligations.*), though instead of separating Stimulus and Agent there were some who would accommodate the Stimulus to the Agent.

3.4. Semantic Frames

In the "case grammar" approach, the role names were primary, and the situations they served to represent were constructed from assemblies of these. What distinguishes frame semantics from case grammar is the switch from taking the role concepts as primary and the situation types as defined in their terms, to taking the frames as primary and defining role distinctions relatively to the frame. This has made it unnecessary to assume a fixed inventory of semantic cases, since the role concepts could be defined relatively to each frame.

4. Frames and the Lexicon

Now if words and other linguistic objects evoke frames, what new responsibilities fall to the linguist and lexicographer? A dictionary designed to recognize that words have background frames will need two-part entries, one part that gives information about the underlying frame, and one part that identifies the place the word in question occupies within that frame.

Familiar dictionaries handle this problem, to some extent, either within the wording of the definition itself, or by referring to some domain the user will need to learn about. No definition of *hypotenuse* would be useful if it failed to mention right angle triangles (Langacker 1987); and words from specialist domains, like baseball or psychoanalysis, are likely to tag definitions with a pointer to the domain itself so that the serious inquirer knows where to look to see how the term fits a larger system: *id, ego, superego, transference*, etc., will have senses that refer to Psychoanalysis and can't really be understood without some knowledge of the whole program; words like *inning, short porch, gap hitter*, etc refer to aspects and strategies of the game of Baseball and will be unintelligible to readers who know nothing about the game. The dictionary user will have to learn about the identified domain in order to approach an understanding of the meaning of the word.

The kind of frame-based dictionary currently being built on the FrameNet project in Berkeley is one which takes for granted that all words, not just specialist words, have meanings that are motivated by some underlying frame. In many cases the frame is so thoroughly and implicitly understood that it doesn't need to be identified in a dictionary intended for native speakers. For example, experience with gravity, rather than an explanation of gravity and its effects, can be part of what is necessary for humans to understand words like *up, down, high, low, tall, short, rise, fall, ascend* and *descend*, but a formal system that needs to be capable of drawing precise inferences cannot learn of verticality by experience in the world.

A frame-informed dictionary should make it possible to look up a word, find adequate information about the requisite background frame, find examples that are analyzed with respect to the way in which the word and its context indicate or presuppose aspects of the frame, learn precisely how the word fits into the frame, and examine other words and phrases that evoke the same frame, to make it possible to understand why one word from the frame was chosen rather than another one.

Dictionary users cannot rely on the definitions as the only source of information about the meanings of words. Consider for example some common definitions of the noun *pedestrian*.

(4) a. a person walking along a road or a developed area
(4) b. a person who travels on foot. (W3)
(4) c. someone who travels by foot

In the WordNet database, synonyms of *pedestrian* are *walker* and *footer*; the hypernym is *traveller*; and the long list of hyponyms includes *stalker*, *rambler* and *tramp*. Something is missing.

From such definitions a person would not be able to "computer" the most natural interpretation of an utterance like (5).

(5) On my way to work this morning I hit a pedestrian.

In one possible envisionment of the situation described by (5), licensed by the definitions just seen, two people are walking somewhere side by side, and one of them gives the other a punch in the face.

A frame-informed definition would have to provide an explanation something like this: In urban settings there are spaces *shared competitively* by people moving on foot and vehicles-—buses, cars, bicycles, etc. *Within that context*, the word designating the people moving about on foot is *pedestrian*. In Langacker's useful terminology, against that *base*, the word *profiles* the one who is on foot. Only by having such a background can we make sense of expressions like *pedestrian zone, pedestrian safety, pedestrian crossing*, and the like. The person who uttered sentence (5) had to have been driving a vehicle, and it was the vehicle that struck a pedestrian.

Definitions of the Word *customer* tend to look like this:

(6) a. a person or organization that buys goods or services from a store or business
(6) b. one that purchases a commodity or service
(6) c. someone who pays for goods or services

Those definitions cover just about everyone we know over age four. This time, suppose we overheard someone say sentence (7).

(7) John tends to be rude to customers.

Clearly, nobody would use this utterance as a way of saying that John is rude to anybody who has ever bought anything. A frame-informed definition would say that in the context of a relationship between a business or company and the people who buy (or might buy) what they offer, the latter are designated *customers*. Having learned from sentence (7) that]ohn was *rude to customers*, we instantly conclude that John works in a company or shop, and the objects of his rudeness are real or potential buyers of his company's products or services.

These have been examples of role-naming nouns, where the noun evokes a frame and profiles the occupant of a role within that frame. In a frame-based lexicon-building project I direct called FrameNet (http://framenet.icsi.berkeley.edu, see Fontenelle 2003) we link words to the frames they evoke, offer explanations of the nature of those frames and the entities or states that characterize the situation types known by these frames, and describe words according to how such information is expressed in sentences containing those words. Most of the items treated so far in FrameNet have been predicators, mostly verbs.

One of these frames dealt with in FrameNet is Revenge, based on situations in which person A commits an offense that injures person B, and person C (who might be identical to B) acts to punish A for the offense. Relevant events and entities to talk about in instances of this frame include the Offender, the Injury, the Injured_party, the Avenger, and the Punishment. For any instance of the Revenge frame, each of these categories is relevant, whether given expression in the associated text or not. The FrameNet treatment of the words in such frames is to extract samples of sentences using them from text corpora, to annotate those sentences with respect to the ways in which the frame's components are expressed in these sentences, and then to prepare descriptions of the combinatory possibilities of the words in both syntactic and semantic terms. Thus, in a sentence like (8), we find the Avenger expressed as the verb's subject, the Offense expressed as the direct object and the Punishment indicated with a VP introduced by In sentence (9), with a different verb, we have the Offender as the direct object, and the Offense and the Punishment unmentioned.

(8) He *avenged* his brother's murder by setting fire to the enemy's village.
(9) He *retaliated* against the enemy.

FrameNet has described approximately 800 frames and 10,000 lexical units that evoke and are explained by the frames. The FrameNet database provides lexical descriptions, frame descriptions, and a large collection of annotated sentences (currently about 150,000) on the basis of which the descriptions have been constructed.

5. Summary

The evolution of the uses of frame within linguistics-—I should say within American linguistics--follows the course of one strand of development in linguistic practice and theory. In the purely "descriptivist" period, the notion of explanation did not play a role: finding namable classes of words or sounds by precise distributional means, or "operational definitions," was the way to know what you are talking about, something that did not rely on imprecise traditions. In the generative period, syntax was central, and the classes of words and phrases worth talking about were precisely those that entered into the basic account of the organization of the language: the word classes identified for these purposes did not need to be "defined" in any traditional way; their validity stood or fell according to the performance of the grammatical theory itself. In the "case grammar" period (here I cannot claim to be talking about the history of linguistics, but of a stream which I followed) the defining contexts were not exclusively matters of syntactic form, but included, for predicators, notions of semantic roles. The final leap inverted the reductionist view that complex predications were made up of structures of roles, but were holistic structures of their own —the frames— and the "roles" had to be understood relatively to the wholes in which they participated. On this meaning, the frames merge with any kind of knowledge that can be evoked by lexical or other linguistic means. Whether in the end there is a principled distinction between cognitive frames and linguistic frames, as conceptual entities, in a world in which anything at all can be a target of human intellection, and discourse, remains an open question.

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